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AI CLINIC

AI for Healthcare Compliance and Regulations Across Countries

Interactive Dashboard and Comparative Regulatory Analysis

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Abstract

Artificial intelligence (AI) is reshaping healthcare through diagnostic imaging, digital pathology, genomic interpretation, drug-discovery support, and clinical decision support. These capabilities create substantial regulatory complexity: software may qualify as Software as a Medical Device (SaMD), process sensitive health data under divergent privacy regimes, and require lifecycle governance when algorithms adapt over time.

This project delivers a research and engineering artefact comprising (1) a structured literature review of AI healthcare compliance across 20 countries in six regions; (2) a curated dataset scoring seven regulatory themes on a 1–10 scale; and (3) an interactive dashboard for geospatial comparison, theme analysis, global trends, use-case readiness, and literature browsing.

Key findings include convergence toward risk-based SaMD classification and comprehensive data protection modelled on the GDPR, persistent disparities in enforcement capacity between advanced and emerging economies, and growing emphasis on post-market surveillance and liability for adaptive AI. The EU AI Act is a watershed in binding AI governance. Clinical adoption in radiology, pathology, genomics, and public-health surveillance amplifies the need for transparent validation, representative datasets, and Total Product Lifecycle oversight.

Keywords: AI healthcare regulation; SaMD; compliance dashboard; data privacy; clinical validation; EU AI Act; post-market surveillance.

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1. Introduction

1.1 General context

The integration of AI into healthcare is one of the most consequential technological shifts of the twenty-first century. AI-based SaMD now assists clinicians in radiology, pathology, ophthalmology, cardiology, and genomics, among other specialties [1], [2]. Major regulators have authorised hundreds of AI/ML-enabled devices, reflecting exponential growth in the field [3], [4].

Unlike traditional medical devices, AI systems can be adaptive, opaque, and dependent on training data that may embed biases—raising regulatory questions that existing frameworks were not designed to address [5], [6]. Ensuring patient safety, data privacy, algorithmic fairness, and accountability therefore requires jurisdiction-specific compliance strategies and continuous monitoring after deployment.

1.2 Problem statement

Regulatory diversity across countries creates practical and scholarly difficulties for four groups of stakeholders:

- **Developers** seeking international market access and harmonised evidence packages;
- **Researchers** studying comparative AI governance;
- **Policymakers** balancing innovation incentives with safety protections;
- **Clinicians and hospital administrators** evaluating whether AI tools meet local legal and ethical requirements.

Public information on AI healthcare regulation is abundant but fragmented: statutes, guidance, ethics principles, and device-authorisation lists sit in separate institutional silos. There is a need for a structured, comparative analysis that is both scholarly (literature synthesis, explicit scoring method) and usable (an interactive dashboard). This project addresses that gap by combining a coded 20-country dataset with a visual analytics interface.

1.2.1 Project objectives

The project aims to:

1. Synthesise peer-reviewed and policy literature on AI healthcare regulation across 20 countries;
2. Define seven thematic dimensions of compliance and score jurisdictional maturity on a 1–10 scale;
3. Build a curated, machine-readable dataset with regional metadata and global trends;
4. Implement an interactive dashboard for exploration, comparison, and literature browsing;
5. Document methodology, architecture, and findings in this report.

1.2.2 Scope and limitations

The analysis covers 20 countries across North America, Europe, East Asia, South/Southeast Asia, the Middle East, Africa, and Oceania. Theme scores are expert-curated syntheses informed by regulatory documents and literature—they are comparative indicators, not official government ratings. Use-case readiness scores in the dashboard are *derived* from theme weights rather than separate regulatory datasets. This report is an academic and decision-support artefact, not legal advice.

1.3 Literature review

This section synthesises regulatory frameworks, privacy regimes, clinical-evidence expectations, ethics, surveillance, and liability for AI in healthcare, drawing on sources from 2018–2026.

1.3.1 Regulatory frameworks for AI medical devices

The IMDRF framework (2014) provides foundational categorisation for software that meets the definition of a medical device without being part of hardware [7]. Major jurisdictions have adopted or adapted it:

- **United States:** the FDA regulates AI/ML SaMD via 510(k), De Novo, and PMA pathways, supplemented by the AI/ML SaMD Action Plan [3];
- **European Union:** the MDR classifies SaMD by intended purpose and risk; the EU AI Act adds high-risk AI obligations [8];
- **Japan:** the PMDA administers the DASH framework for SaMD [9];
- **China:** the NMPA issued guiding principles for AI medical-device registration [10].

A near-universal trend is proportional regulation by potential harm. Higher-risk applications (for example autonomous diagnosis) face stricter pre-market evaluation and post-market obligations [11]. Table 1.1 summarises representative classification systems.

Table 1.1: Risk-based medical-device classification systems (selected jurisdictions)

Region	System	Classes
USA	FDA risk-based	I, II, III
EU	MDR risk-based	I, IIa, IIb, III
China	NMPA three-tier	I, II, III
Canada	Health Canada	I, II, III, IV
Japan	PMDA / PMD Act	I, II, III, IV
Australia	TGA	I, IIa, IIb, III

AI/ML algorithms can evolve with new data. Regulators therefore move from point-in-time approval toward lifecycle approaches [12], [13]. The FDA Predetermined Change Control Plan (PCCP) allows pre-specified modifications without a new submission for every change [14]. The EU AI Act requires ongoing conformity and risk management. Good Machine Learning Practice (GMLP) principles from the FDA, Health Canada, and the MHRA guide development practice [15].

1.3.2 Data privacy and governance

The GDPR (2018) established global benchmarks: lawfulness, purpose limitation, data minimisation, accountability, data-subject rights, and Data Protection Impact Assessments. Article 22 on automated decision-making has significant implications for clinical AI [16]. Comprehensive data-protection laws have since spread to Brazil (LGPD), India (DPDP), China (PIPL), South Africa (POPIA), Nigeria (NDPA), Kenya, Thailand, Gulf states, and Switzerland—with varying GDPR alignment.

The United States relies on HIPAA for Protected Health Information [17]. The EU proposes a European Health Data Space for primary and secondary use. Cross-border genomic and AI validation increasingly depend on lawful transfer mechanisms and representativeness of training data [5].

1.3.3 Clinical validation, safety, and transparency

Clinical-evidence requirements for AI SaMD remain contested. Studies of FDA-cleared devices highlight limitations in external validation and subgroup performance [4]. Algorithmic bias in population-health tools demonstrates real-world harm potential [18], [19].

Transparency debates pit post-hoc explainable AI against interpretable-by-design models [20], [21]. Regulators increasingly expect human-readable documentation, performance monitoring, and clinician-facing intelligibility even when full algorithmic transparency is infeasible.

1.3.4 Ethics and accountability

WHO ethics guidance emphasises human oversight, transparency, accountability, inclusiveness, and sustainability [22]. Jobin et al. mapped hundreds of AI ethics guidelines globally [23]. A persistent “ethics gap” exists between voluntary principles and enforceable practice [24], [25]. Regional perspectives differ: EU rights-based approaches, US innovation-oriented voluntary frameworks, China’s state-guided development, African Ubuntu-influenced discourse, and Middle Eastern intersections of Islamic ethics with modernisation.

1.3.5 Post-market surveillance and liability

Post-market systems include FDA MedWatch, EU MDR serious-incident reporting, and the UK Yellow Card. AI-specific harms (silent degradation, systematic bias) may not fit traditional adverse-event categories [26]. Liability frameworks largely rely on product liability and malpractice law, which may be inadequate for opaque AI [27]. Proposed EU AI-liability instruments would shift causation burdens in some cases.

1.3.6 Comparative regional synthesis

North America combines advanced agencies (FDA, Health Canada) with fragmented US federal AI law. **Europe** leads with the EU AI Act, MDR, and GDPR integration. **East Asia** shows rapid approval activity and algorithm-specific rules (China, South Korea). **South/Southeast Asia** spans Singapore’s governance leadership and India’s emerging digital-health infrastructure. **Middle East** pairs Gulf investment with Israel’s innovation ecosystem. **Africa** faces capacity constraints but benefits from mobile-health innovation and new data-protection laws.

2. Methodology

2.1 Research design

The project follows a mixed documentary and engineering design. First, a structured literature and policy review produced a common coding schema for 20 jurisdictions. Second, each jurisdiction was scored on seven compliance themes. Third, the coded dataset was implemented as an interactive dashboard so that scores, gaps, and narratives can be explored without reading the raw JSON. No machine-learning model generates scores at runtime; all scores are curated from documentary evidence.

2.2 Literature search strategy

A structured review used PubMed, Scopus, IEEE Xplore, Google Scholar, and SSRN, combined with regulatory sources (FDA, EMA, MHRA, NMPA, PMDA, TGA, Health Canada, WHO, OECD, IMDRF) and national AI strategies. Search terms included “AI healthcare regulation,” “SaMD compliance,” “health data governance,” and “algorithmic transparency medicine.” Inclusion criteria: publications from 2018–2026 addressing AI regulation, compliance, or ethics in healthcare. Thirty-two peer-reviewed and policy references inform the synthesis in Section 1.3.

2.3 Country sample and coding schema

Twenty jurisdictions were selected to cover North America, Europe, Asia, the Middle East, Africa, and Oceania, including both advanced regulators and emerging markets. Each country record uses the same fields: regulatory body, privacy law, AI-specific regulation, medical-device framework, approval process, data governance, clinical validation, transparency, ethics, post-market surveillance, liability, legislations, challenges, and notable developments.

2.4 Theme scoring

Each country is scored on seven regulatory themes on a 1–10 ordinal scale. A score of 1 means little or no documented requirement for that theme; 10 means a mature, enforceable framework with clear guidance and operational practice. Documentary evidence was mapped to scores using four criteria: presence of law or guidance; specificity to health AI / SaMD; enforcement or reporting mechanisms; and alignment with IMDRF / WHO / GMLP.

Table 2.1: Compliance theme definitions

Theme	Description
Data Privacy	Health-data protection laws, cross-border transfer rules, DPIAs
Clinical Validation	Evidence standards, clinical evaluation, bias assessment
Approval Process	SaMD pathways, AI-specific guidance, regulatory sandboxes
Transparency	Explainability expectations, documentation, clinician intelligibility
Ethics	National AI ethics frameworks, fairness, human oversight
Post-Market	Adverse-event reporting, lifecycle monitoring, update governance
Liability	Product liability, malpractice interfaces, AI-specific proposals

The overall compliance score shown in dashboard KPIs, the world map, and top-performer

charts is the arithmetic mean of the theme scores currently selected in the slicers:

$$\text{Overall score} = \frac{1}{k} \sum_{i=1}^k s_i \quad (2.1)$$

where s_i is the score of selected theme i and k is the number of selected themes, rounded to one decimal place. Changing theme filters recalculates this average immediately.

A *compliance gap* for a theme is $\max(s) - \min(s)$ among countries in the current filter. A large gap means some jurisdictions have strong rules while others have almost none.

2.5 Maturity classification

Maturity is a country-level qualitative class stored in the dataset (`maturity_level`). It is assigned from documentary evidence—AI-specific rules, device pathway, approval volume, and privacy/ethics enforcement—not from a hidden formula on the scores alone. Classes are:

- **Early** — limited AI-specific healthcare regulation, few or no documented AI device approvals, and basic or recently introduced data-protection rules (e.g. Nigeria, Kenya);
- **Emerging** — data protection and/or voluntary AI ethics exist, but SaMD / AI-device pathways are still forming (e.g. Saudi Arabia, South Africa);
- **Developing** — a medical-device framework is in place and AI guidance is appearing, with moderate institutional capacity (e.g. India, Brazil);
- **Moderate** — functioning SaMD registration, international alignment, and a working post-market system (e.g. Australia);
- **Advanced** — dedicated AI/ML SaMD pathways or high-risk AI law, substantial device authorisations, and lifecycle governance (e.g. United States, EU, United Kingdom, Germany, Japan, Singapore).

Device risk classes (FDA I–III, MDR I–III, etc.) describe the *product*. Maturity classes describe the *jurisdiction's* governance. They are independent constructs.

2.6 Use-case readiness model

Use-case readiness (radiology, pathology, genomics, drug discovery) is a weighted mean of the same theme scores, using weights that reflect which themes matter most for that clinical application. For example, radiology readiness weights clinical validation and post-market surveillance heavily; genomics emphasises data privacy; drug discovery emphasises approval process and transparency. This links regulatory maturity to deployment contexts while keeping the derivation explicit.

2.7 Dashboard implementation

The interactive artefact is a React 18 application (Vite, TypeScript, Tailwind CSS, Recharts, react-simple-maps). It loads `compliance_dataset.json`, applies region / maturity / theme filters, and computes overall averages, gaps, use-case readiness, and charts. Pages cover overview, comparison, theme analysis, trends, country details, and the literature review.

A Streamlit prototype (`app.py`) remains in the repository as a reference implementation. The public deliverable is the web dashboard, deployed to GitHub Pages. The canonical dataset is version-controlled in `data/compliance_dataset.json`.

Table 2.2: Implementation stack

Component	Technology
Public dashboard	React 18, Vite, TypeScript, Tailwind CSS
Charts / map	Recharts, react-simple-maps
Reference prototype	Python, Streamlit, Plotly, Pandas
Data store	JSON (version-controlled)
Hosting	GitHub Pages

2.8 Pipeline summary

1. Scope 20 jurisdictions across six regions.
2. Collect primary regulatory texts and secondary literature (2018–2026).
3. Code each country into a common schema.
4. Assign 1–10 theme scores and a maturity label.
5. Derive overall scores, gaps, and use-case readiness in the dashboard.
6. Document findings in this report.

3. Results

This chapter presents the comparative findings from the 20-jurisdiction dataset. We first report overall rankings and theme-level patterns with figures, then analyse regional clusters, quantify inter-country gaps, profile every jurisdiction, and finally link the scores to clinical use cases.

3.1 Overall ranking and score matrix

Table 3.1 ranks all jurisdictions by the arithmetic mean of the seven theme scores. The European Union aggregate, Germany, and the United States occupy the top tier (overall ≥ 7.6). Kenya and Nigeria sit at the bottom of the sample (overall ≈ 2.7 – 2.9), illustrating the capacity gap between advanced regulators and early-stage systems.

Table 3.1: Overall ranking of jurisdictions by mean theme score

Rank	Country	Region	Maturity	Overall	Devices
1	European Union	Europe	Advanced	9.1	600
2	Germany	Europe	Advanced	8.7	180
3	United States	North America	Advanced	7.6	950
4	United Kingdom	Europe	Advanced	7.1	200
5	Canada	North America	Advanced	7.1	120
6	Japan	Asia	Advanced	7.0	150
7	Singapore	Asia	Advanced	6.9	60
8	Switzerland	Europe	Advanced	6.9	90
9	China	Asia	Advanced	6.7	300
10	Australia	Oceania	Moderate	6.6	80
11	South Korea	Asia	Advanced	6.6	200
12	Israel	Middle East	Advanced	6.3	100
13	Brazil	South America	Developing	5.3	40
14	UAE	Middle East	Emerging	4.6	35
15	Saudi Arabia	Middle East	Emerging	4.4	25
16	India	Asia	Developing	4.3	30
17	Thailand	Asia	Developing	3.9	15
18	South Africa	Africa	Emerging	3.6	10
19	Kenya	Africa	Early	2.7	3
20	Nigeria	Africa	Early	2.6	5

Figure 3.1 visualises the same ranking as a horizontal bar chart coloured by region. The visual ordering makes three patterns immediate: (i) Europe and North America dominate the upper half; (ii) Asia is heterogeneous, stretching from Japan/Singapore near the top-middle to Thailand/India in the lower-middle; (iii) Africa occupies the lower tail, with South Africa ahead of Kenya and Nigeria.

Table 3.2 provides the full theme-by-theme matrix used by the dashboard. Privacy = Data Privacy; Clin. = Clinical Validation; Appr. = Approval Process; Trans. = Transparency; Eth. = Ethics; Post = Post-Market; Liab. = Liability; Devices = reported AI device approvals (indicative counts).

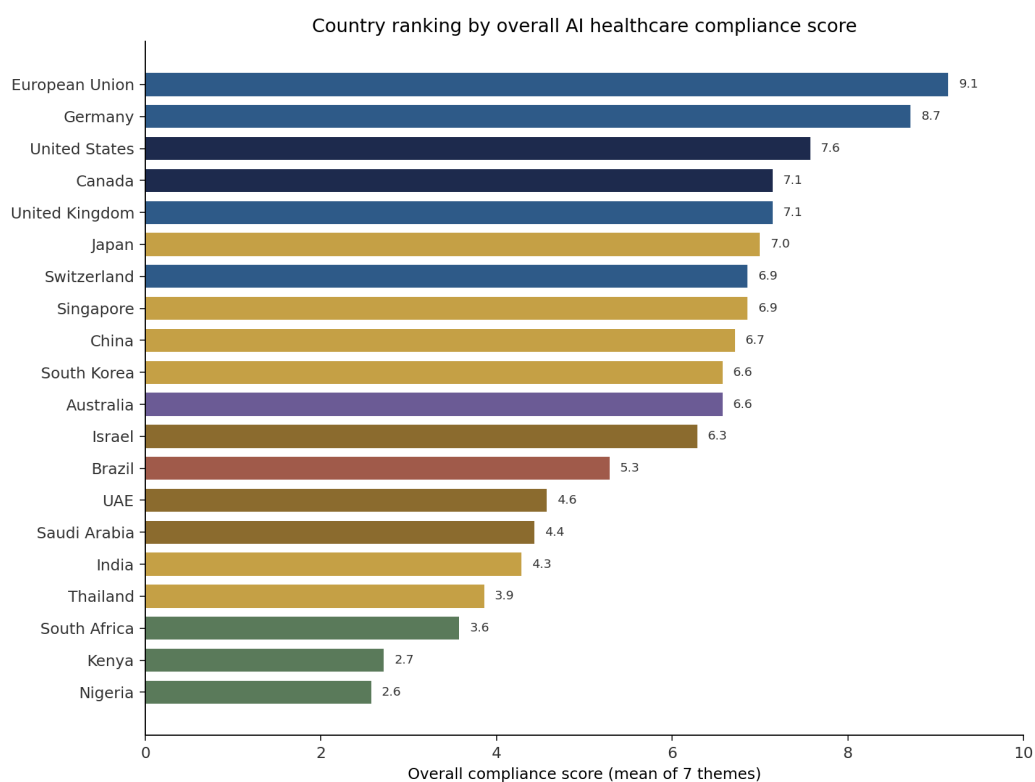


Figure 3.1: **Overall country ranking.** Each bar is the mean of the seven theme scores (1–10) for one jurisdiction. Bars are coloured by region. Longer bars mean stronger documented AI healthcare compliance. Read from bottom (highest) to top (lowest): the EU, Germany, and the United States lead; Kenya and Nigeria trail. Use this figure to show the jury the global spread of regulatory maturity at a glance.

Table 3.2: Country compliance scores (1–10) across seven themes

Country	Region	Maturity	Priv.	Clin.	Appr.	Trans.	Eth.	Post	Liab.	Devices
European Union	Europe	Advanced	10	9	9	9	10	9	8	600
Germany	Europe	Advanced	10	9	9	8	9	9	7	180
United States	North America	Advanced	8	9	9	6	6	8	7	950
United Kingdom	Europe	Advanced	8	8	7	7	7	7	6	200
Canada	North America	Advanced	7	8	8	7	7	7	6	120
Japan	Asia	Advanced	7	8	8	6	7	8	5	150
Singapore	Asia	Advanced	7	7	7	8	8	6	5	60
Switzerland	Europe	Advanced	8	8	8	6	6	7	5	90
China	Asia	Advanced	7	7	7	7	6	7	6	300
Australia	Oceania	Moderate	7	7	7	6	7	7	5	80
South Korea	Asia	Advanced	7	7	8	6	6	7	5	200
Israel	Middle East	Advanced	6	8	7	6	6	6	5	100
Brazil	South America	Developing	7	5	5	5	5	5	5	40
UAE	Middle East	Emerging	5	5	5	5	5	4	3	35
Saudi Arabia	Middle East	Emerging	5	4	5	5	5	4	3	25
India	Asia	Developing	5	4	4	4	5	4	4	30
Thailand	Asia	Developing	5	4	4	4	4	3	3	15
South Africa	Africa	Emerging	6	3	3	3	4	3	3	10
Kenya	Africa	Early	5	2	3	2	3	2	2	3
Nigeria	Africa	Early	4	2	3	2	3	2	2	5

3.2 Theme-level comparison

Figures 3.2 and 3.3 and Table 3.3 answer two questions: which themes are strongest worldwide on average, and where countries diverge the most. A large gap means a divergence zone—some jurisdictions already have detailed AI rules, while others have almost none. Clinical validation and transparency show particularly wide spreads, which matters for multi-country evidence packages.

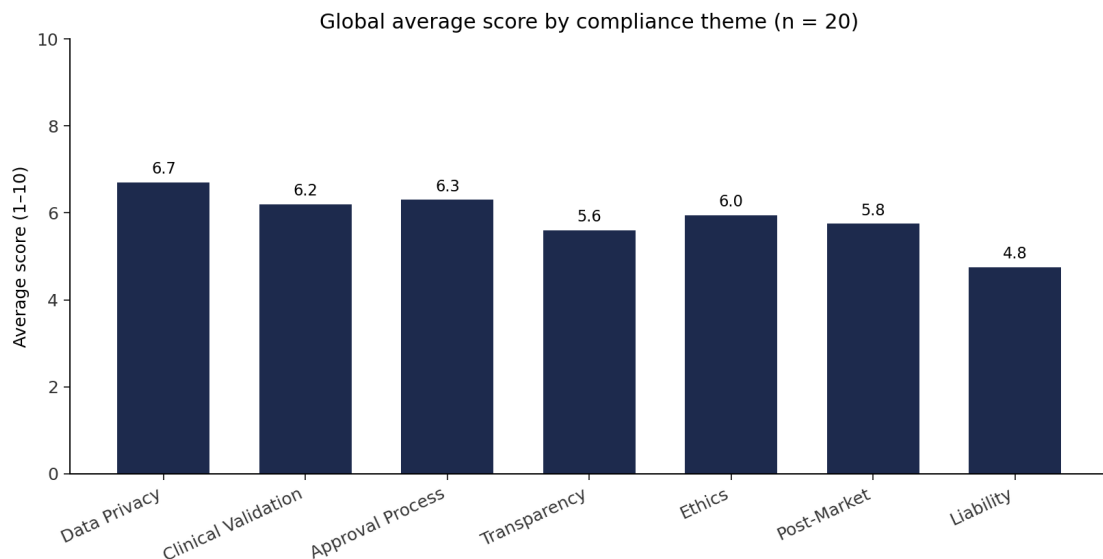


Figure 3.2: **Global average score by theme.** For each of the seven compliance themes, the bar shows the mean score across all 20 countries. Higher bars mean that theme is more mature worldwide. Approval process and data privacy tend to lead (device pathways and post-GDPR privacy laws are widespread). Post-market surveillance and liability tend to lag, showing that lifecycle governance and accountability remain the weakest global pillars.

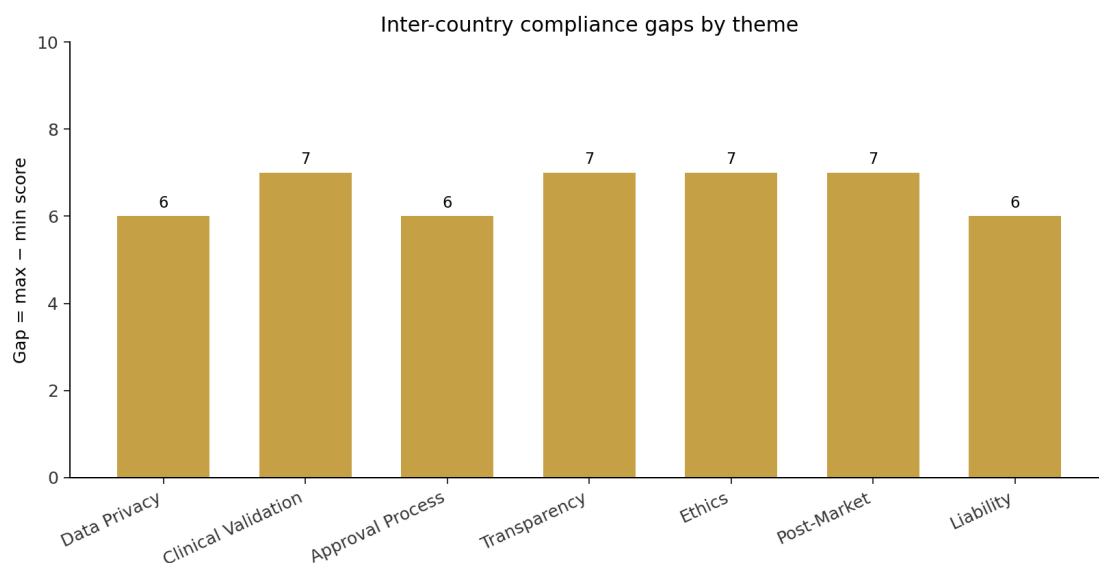


Figure 3.3: **Inter-country gaps by theme.** Gap = maximum country score minus minimum country score for that theme. A large gap means strong divergence: some countries have detailed AI rules while others have almost none. Clinical validation and transparency typically show the widest gaps—important for developers who need multi-country evidence packages.

Table 3.3: Theme leaders and laggards across the 20-country sample

Theme	Gap	Leader	Laggard
Data Privacy	6	European Union	Nigeria
Clinical Validation	7	United States	Nigeria
Approval Process	6	United States	South Africa
Transparency	7	European Union	Nigeria
Ethics	7	European Union	Nigeria
Post-Market	7	European Union	Nigeria
Liability	6	European Union	Nigeria

3.2.1 How to read the theme leaders and laggards

Table 3.3 lists, for each theme, the highest-scoring jurisdiction (leader), the lowest-scoring jurisdiction (laggard), and the numeric gap between them. The pattern is consistent and easy to summarise:

- **Europe leads on rights and accountability themes.** The EU (and European peers) top data privacy, transparency, ethics, post-market, and liability. That reflects binding instruments—notably GDPR and the EU AI Act—rather than voluntary ethics checklists alone [8].
- **The United States leads on market-access themes.** The US tops clinical validation and approval process, driven by FDA SaMD pathways, published authorisation counts, and related clinical evidence practice [3], [4].
- **Early African jurisdictions dominate the laggard column.** Nigeria (and South Africa on approval process) appear repeatedly at the bottom. Device pathways and AI-specific adverse-event infrastructure are still forming there, so scores stay low even when privacy statutes exist.

In short: advanced regulators pull ahead where either enforceable privacy/ethics law or deep device pathways already exist; the largest gaps appear where those institutions are still early.

3.3 Maturity distribution

Figure 3.4 shows the frequency of maturity classes. Most jurisdictions in the sample are labelled **Advanced**, reflecting deliberate selection of major regulators alongside emerging markets. The smaller Early / Emerging / Developing groups are analytically important: they define the lower bound against which gaps are measured and they are the primary audience for capacity-building recommendations. Figure 3.5 summarises mean overall scores by region.

Maturity is not a hidden transform of the overall score. For example, China and Israel are Advanced despite mid-range privacy or liability scores, because they operate dedicated AI/device pathways with non-trivial approval activity. Australia is Moderate despite solid theme averages, reflecting a functioning but less AI-specialised pathway relative to FDA/EU peers.

3.4 Regional comparative analysis

Table 3.4 and Figures 3.5–3.6 summarise regional means. Europe leads on the composite, followed by North America and Oceania; Africa trails substantially. Asia’s mean sits in the middle because high performers (Japan, Singapore, South Korea, China) are averaged with developing pathways (India, Thailand).

Table 3.4: Regional mean scores and sample size

Region	n	Overall	Priv.	Clin.	Appr.	Trans.	Eth.	Post	Liab.
Africa	3	3.0	5.0	2.3	3.0	2.3	3.3	2.3	2.3
Asia	6	5.9	6.3	6.2	6.3	5.8	6.0	5.8	4.7
Europe	4	8.0	9.0	8.5	8.2	7.5	8.0	8.0	6.5
Middle East	3	5.1	5.3	5.7	5.7	5.3	5.3	4.7	3.7
North America	2	7.4	7.5	8.5	8.5	6.5	6.5	7.5	6.5
Oceania	1	6.6	7.0	7.0	7.0	6.0	7.0	7.0	5.0
South America	1	5.3	7.0	5.0	5.0	5.0	5.0	5.0	5.0

3.4.1 North America

The United States and Canada are both Advanced. The US leads on approval process and clinical validation (FDA pathways plus PCCP) but is weaker on binding transparency and

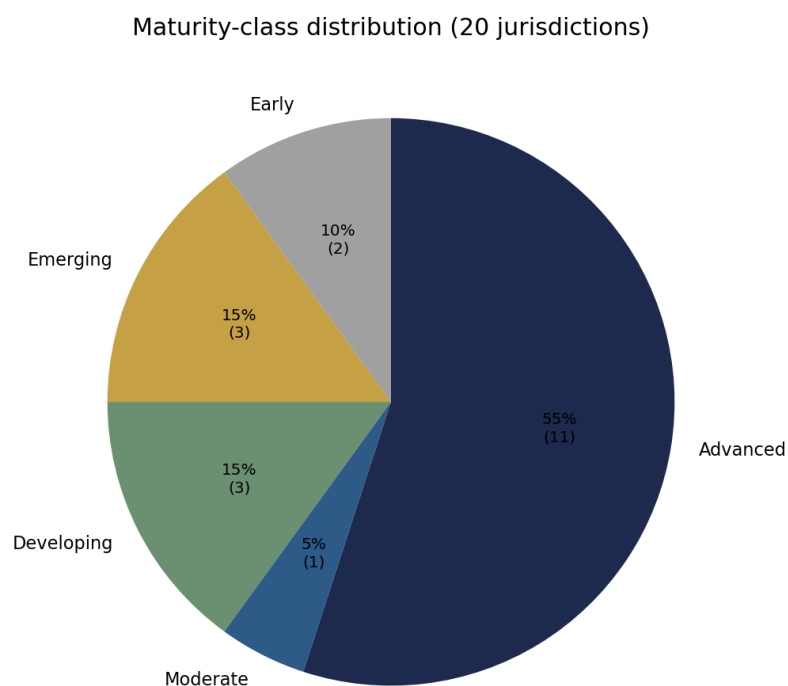


Figure 3.4: **Maturity-class distribution.** Pie slices show how many of the 20 jurisdictions fall into each qualitative maturity class (Early, Emerging, Developing, Moderate, Advanced). Maturity is assigned from documentary evidence (AI-specific rules, device pathway, approval activity, privacy/ethics enforcement)—not from a hidden formula on the scores alone. The Advanced slice is large because the sample deliberately includes major regulators; Early/Emerging countries define the capacity gap.

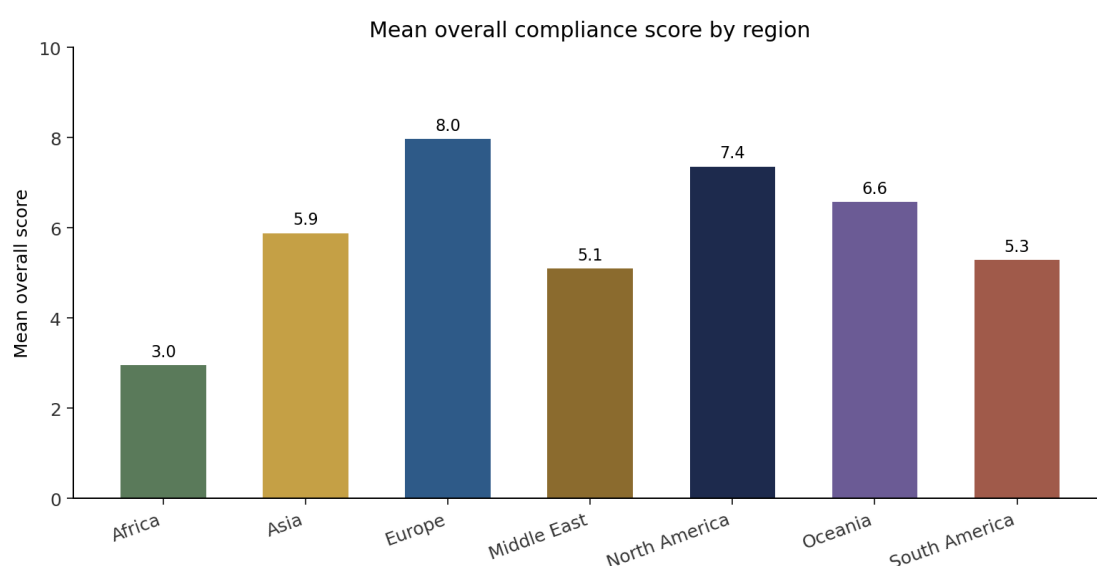


Figure 3.5: **Mean overall compliance score by region.** Each bar is the average of country overall scores inside that region. Europe leads, followed by North America and Oceania; Africa is lowest. Asia sits in the middle because high performers (Japan, Singapore, South Korea, China) are averaged with developing pathways (India, Thailand). Oceania and South America rest on single-country observations (Australia, Brazil) and should not be over-generalised.

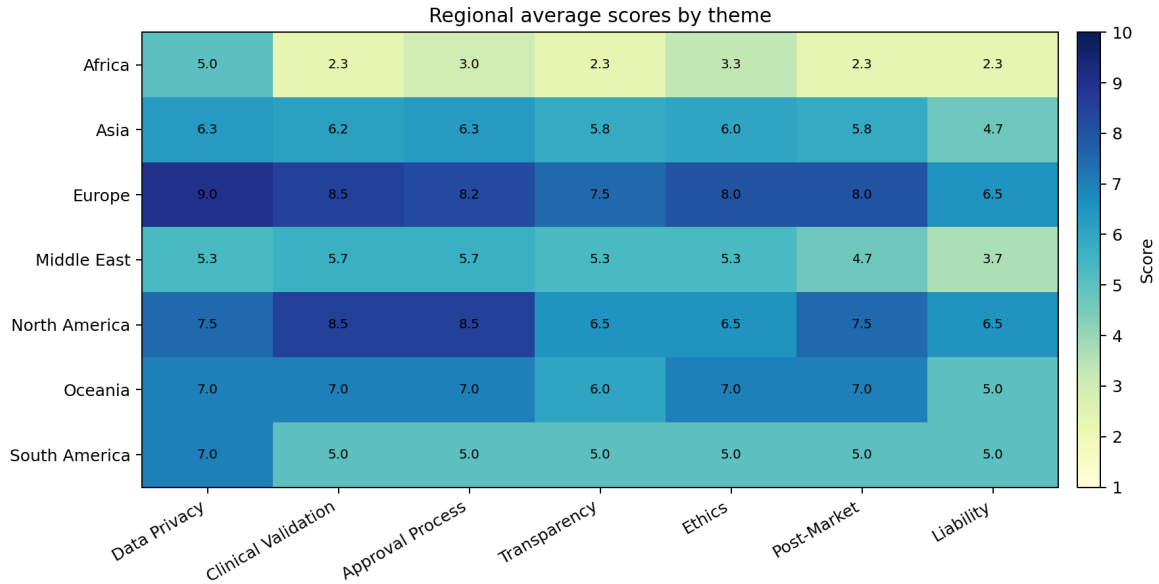


Figure 3.6: **Regional theme heatmap.** Rows are regions; columns are the seven compliance themes. Cell colour and number show the regional mean score (darker = higher). Read across a row to see a region’s strengths/weaknesses; read down a column to see which region leads a theme. Europe is strong across privacy/ethics; North America is strong on clinical validation and approval; Africa shows low values on most AI-specific themes despite privacy progress in some countries.

federal ethics law; HIPAA is sectoral rather than GDPR-equivalent [17]. Canada pairs Health Canada device classes with closer GMLP alignment. North America offers high market-access volume with careful multi-state privacy mapping.

3.4.2 Europe

The EU aggregate, Germany, the United Kingdom, and Switzerland form the densest high-compliance cluster. The EU AI Act adds a binding high-risk layer on MDR SaMD and GDPR, lifting ethics and transparency relative to voluntary NIST-style frameworks [8]. Europe is the reference region for rights-based AI healthcare governance.

3.4.3 Asia

Asia is the most internally diverse region. Japan and Singapore combine mature agencies with structured AI guidance; South Korea and China show high approval throughput and algorithm-specific rules [9], [10]. India and Thailand remain Developing: medical-device frameworks exist, but AI-specific validation and post-market duties are thinner. Cross-border AI deployment in Asia cannot assume a single evidence package.

3.4.4 Middle East, Africa, Oceania, and South America

Israel is Advanced (innovation- and device-driven). Saudi Arabia and the UAE are Emerging (strategy-led catch-up). In Africa, South Africa (POPIA, Emerging) outranks Kenya and Nigeria (Early): data-protection progress outpaces AI-device infrastructure. Australia (Moderate) offers a functioning TGA pathway; Brazil (Developing) is privacy-strong (LGPD) relative to clinical-validation scores.

3.5 Peer jurisdiction comparison

Figure 3.7 compares five representative jurisdictions side by side across all seven themes: the United States and the European Union (Advanced leaders), China (Advanced Asian regulator with high approval activity), India (Developing), and South Africa (Emerging). Figure 3.8 complements this by showing radar shapes for the five highest overall scorers.

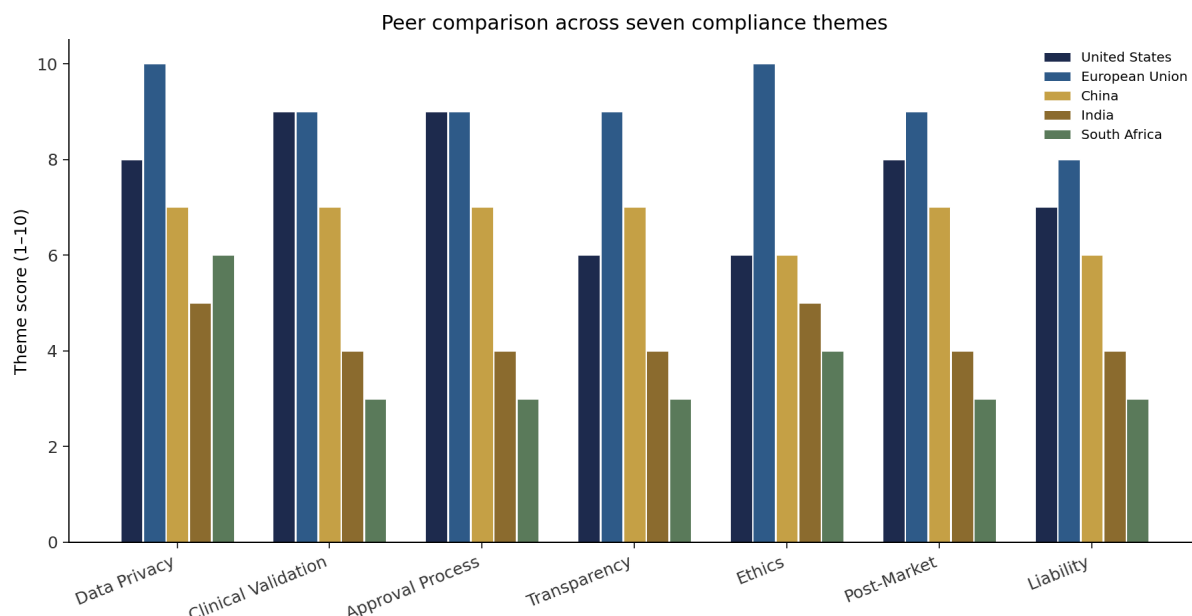


Figure 3.7: **Peer comparison across seven themes.** Grouped bars compare the United States, European Union, China, India, and South Africa on each compliance theme (score 1–10). Read theme by theme: the US and EU are high on most axes, with the EU typically stronger on privacy/ethics/transparency and the US strong on clinical validation and approval. China is competitive on approval and mid-range on privacy. India and South Africa sit lower, especially on clinical validation, post-market, and liability—illustrating the Advanced vs Developing/Emerging gap in one chart.

3.6 Pairwise and strategic comparisons

United States vs. European Union: Both are Advanced. The US offers speed and volume through established SaMD pathways and PCCP for adaptive algorithms [14]. The EU offers stricter enforceable transparency, ethics, and privacy duties under GDPR + AI Act + MDR. A dual-track strategy (US clinical evidence + EU conformity/quality system) remains common for global SaMD vendors [11].

China vs. Japan: Both are Advanced Asian regulators with distinct emphases: China emphasises registration guidance and domestic algorithm filing practices; Japan emphasises PMDA DASH evaluation and quality-system discipline. Privacy scores are similar; liability remains a shared soft spot relative to Europe.

Singapore vs. India: Singapore’s Advanced governance (clear guidance, sandbox culture, strong ethics/transparency relative to region) contrasts with India’s Developing trajectory (large digital-health ambition, DPDP privacy progress, thinner AI-device clinical pathway). For ASEAN-oriented products, Singapore is often the regional reference; India is a scale market requiring staged compliance investment.

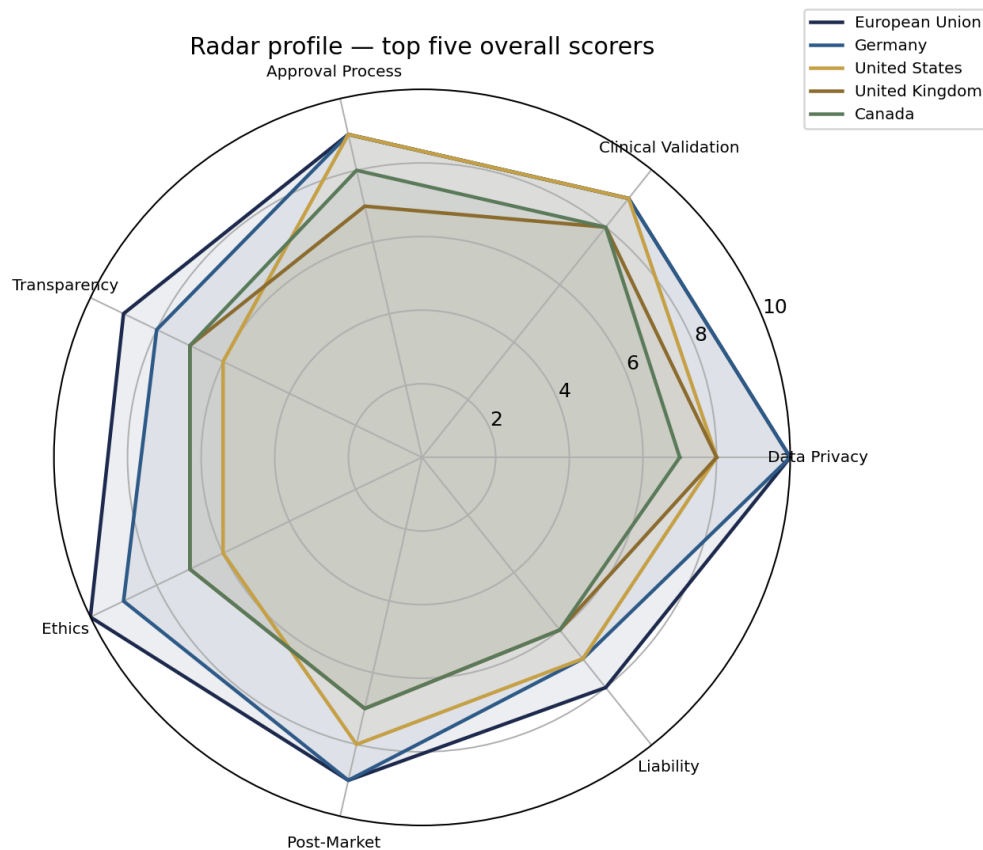


Figure 3.8: **Radar profiles of the top five scorers;** Each coloured polygon is one top jurisdiction across the seven themes (axes from centre 0 to outer rim 10). A wider polygon means stronger scores. Compare shapes: EU/Germany typically extend farthest on privacy and ethics; the United States extends farthest on clinical validation and approval process, with a relative dip on transparency/ethics. This shows that “best country” depends on which theme matters for the product.

Israel vs. Gulf states: Israel’s Advanced profile is innovation- and device-driven. Saudi Arabia and the UAE are investment- and strategy-driven Emerging systems: scores are mid-low today, but national AI agendas imply rapid movement if pathways and surveillance infrastructure catch up.

South Africa vs. Kenya / Nigeria: Within Africa, South Africa’s POPIA-era privacy and relatively stronger institutions produce a clearer Emerging profile. Kenya and Nigeria remain Early on clinical validation, transparency, and post-market themes despite recent data-protection statutes—illustrating that privacy law alone does not raise overall healthcare-AI compliance.

3.7 Country-by-country profiles

The following compact profiles, ordered by overall score, summarise each jurisdiction. They are the narrative counterpart of the dashboard Country Details page and should be read with Tables 3.1 and 3.2.

European Union Europe; Advanced; overall 9.1/10; European Medicines Agency (EMA), National Competent Authorities, Notified Bodies; 600 devices; first AI instrument 2018. Scores: P10 C9 A9 T9 E10 PM9 L8. Privacy: GDPR (General Data Protection Regulation). AI rule: EU AI Act (2024). Challenges: Complex multi-layered regulatory landscape. Harmonization across 27 member states. Notified Body capacity constraints. Risk of over-regulation stifling innov... Notable: EU AI Act is the world’s first comprehensive AI law. High-risk healthcare AI must comply by 2026-2027. EHDS aims to create a unified health data framework.

Germany Europe; Advanced; overall 8.7/10; BfArM (Federal Institute for Drugs and Medical Devices); 180 devices; first AI instrument 2018. Scores: P10 C9 A9 T8 E9 PM9 L7. Privacy: GDPR + BDSG (Federal Data Protection Act). AI rule: EU AI Act (as EU member). National AI Strategy (2018, updated 2020).. Challenges: Historically strict data protection culture can slow health data utilization. DiGA program growing pains. Interoperability of health IT systems. Notable: DiGA is the world’s first regulatory pathway for prescribed digital therapies reimbursed by statutory health insurance. Health Data Use Act aims to unlock re...

United States North America; Advanced; overall 7.6/10; FDA (Food and Drug Administration); 950 devices; first AI instrument 2017. Scores: P8 C9 A9 T6 E6 PM8 L7. Privacy: HIPAA (Health Insurance Portability and Accountability Act). AI rule: FDA AI/ML-Based SaMD Action Plan (2021). Challenges: Fragmented state-level privacy laws. No comprehensive federal AI legislation. Balancing innovation speed with safety oversight for adaptive algorithms. Notable: FDA has authorized over 950 AI/ML-enabled medical devices as of 2025. Predetermined Change Control Plan allows manufacturers to pre-specify algorithm modific...

United Kingdom Europe; Advanced; overall 7.1/10; MHRA (Medicines and Healthcare products Regulatory Agency); 200 devices; first AI instrument 2018. Scores: P8 C8 A7 T7 E7 PM7 L6. Privacy: UK GDPR & Data Protection Act 2018. AI rule: Pro-innovation AI Regulation Framework (2023). Challenges: Post-Brexit regulatory divergence from EU. Balancing light-touch innovation-friendly approach with patient safety. Transition from CE to UKCA marking. Notable: UK positioned as pro-innovation alternative to EU AI Act. AI Safety Institute is a world-first government body for AI safety research. NHS AI Lab supports AI...

Canada North America; Advanced; overall 7.1/10; Health Canada; 120 devices; first AI instrument 2019. Scores: P7 C8 A8 T7 E7 PM7 L6. Privacy: PIPEDA (Personal Information Protection and Electronic Documents Act), Provincial health privacy laws (e.g., Ontario PHIPA). AI rule: AIDA (Artificial Intelligence and Data Act, proposed in Bill C-27). Challenges: Federal-provincial jurisdiction complexity. AIDA legislation delayed. Keeping pace with rapid AI advancement in healthcare. Notable: Canada co-developed GMLP with FDA and MHRA. Pan-Canadian Health Data Strategy in development. Strong AI research ecosystem (MILA, Vector, Amii) informing pol...

Japan Asia; Advanced; overall 7.0/10; PMDA (Pharmaceuticals and Medical Devices Agency); 150 devices; first AI instrument 2019. Scores: P7 C8 A8 T6 E7 PM8 L5. Privacy: APPI (Act on Protection of Personal Information, amended 2022). AI rule: AI Guidelines for Business (2024), Social Principles of Human-Centric AI (2019). Challenges: Aging society drives demand but conservative regulatory culture. Language barriers

in international harmonization. Slow adoption of digital health relative t... Notable: Japan's regulatory sandbox (IDATEN) accelerates innovative device approval. Society 5.0 vision integrates AI in healthcare transformation. PMDA actively deve...

Singapore Asia; Advanced; overall 6.9/10; HSA (Health Sciences Authority); 60 devices; first AI instrument 2019. Scores: P7 C7 A7 T8 E8 PM6 L5. Privacy: PDPA (Personal Data Protection Act, 2012, amended 2021). AI rule: Model AI Governance Framework (2019, updated 2020), AI Verify (2022). Challenges: Small domestic market limits local clinical evidence generation. Balancing business-friendly environment with robust oversight. Regional influence vs. global... Notable: AI Verify is the world's first AI governance testing framework. Singapore positioned as AI governance thought leader in ASEAN. Strong public-private partners...

Switzerland Europe; Advanced; overall 6.9/10; Swissmedic; 90 devices; first AI instrument 2023. Scores: P8 C8 A8 T6 E6 PM7 L5. Privacy: nFADP (New Federal Act on Data Protection, 2023). AI rule: No AI-specific law. Federal Council AI guidelines. Bilateral alignment with EU standards.. Challenges: Dependency on EU regulatory decisions without EU membership voting rights. Maintaining bilateral agreements post-EU AI Act. Small market size. Notable: Strong pharma/medtech hub (Novartis, Roche). nFADP brings Swiss privacy law to GDPR-equivalent. Active AI health research at ETH Zurich and EPFL.

China Asia; Advanced; overall 6.7/10; NMPA (National Medical Products Administration); 300 devices; first AI instrument 2021. Scores: P7 C7 A7 T7 E6 PM7 L6. Privacy: PIPL (Personal Information Protection Law, 2021). AI rule: Interim Measures for Generative AI (2023), Algorithm Recommendation Regulations (2022), Deep Synthesis Regulations (2023). Challenges: Rapid regulatory evolution creates compliance complexity. Data localization requirements limit international collaboration. Balancing state surveillance prio... Notable: China is among the first to regulate algorithmic recommendations and generative AI. NMPA rapidly approving AI diagnostic tools. National AI standardization e...

Australia Oceania; Moderate; overall 6.6/10; TGA (Therapeutic Goods Administration); 80 devices; first AI instrument 2019. Scores: P7 C7 A7 T6 E7 PM7 L5. Privacy: Privacy Act 1988 (amended), Australian Privacy Principles (APPs). AI rule: Australia's AI Ethics Principles (2019), Voluntary AI Safety Standard (2024). Challenges: Voluntary nature of AI ethics framework limits enforcement. Resource constraints in TGA for AI evaluation. Geographic barriers to clinical trial recruitment. Notable: Australia's AI Ethics Principles among the earliest globally. TGA actively developing SaMD regulatory pathways. National Digital Health Strategy driving AI i...

South Korea Asia; Advanced; overall 6.6/10; MFDS (Ministry of Food and Drug Safety); 200 devices; first AI instrument 2019. Scores: P7 C7 A8 T6 E6 PM7 L5. Privacy: PIPA (Personal Information Protection Act, amended 2023). AI rule: AI Framework Act (proposed), National AI Strategy (2019). Challenges: Balancing rapid technological adoption with regulatory oversight. Harmonizing domestic standards with international frameworks. SME compliance burden. Notable: South Korea has one of the highest AI medical device approval rates globally. MFDS Regulatory Science Center is a dedicated AI device evaluation unit. K-BioH...

Israel Middle East; Advanced; overall 6.3/10; MOH AMAR (Division of Medical Devices, Ministry of Health); 100 devices; first AI instrument 2022. Scores: P6 C8 A7 T6 E6 PM6 L5. Privacy: Privacy Protection Law (1981, amended), Health Information Exchange regulations. AI rule: AI Policy and Ethics Framework (2022), National AI Program. Challenges: Outdated base privacy legislation. Balancing startup-friendly ecosystem with regulatory rigor. Small domestic market necessitates international focus. Notable: Israel is a global health AI innovation hub. Unique national health data infrastructure enables world-class AI research. Highest health AI startups per capit...

Brazil South America; Developing; overall 5.3/10; ANVISA (Agência Nacional de Vigilância Sanitária); 40 devices; first AI instrument 2020. Scores: P7 C5 A5 T5 E5 PM5 L5. Privacy: LGPD (Lei Geral de Proteção de Dados, 2020). AI rule: AI Bill (PL 2338/2023, under legislative process). Challenges: AI regulatory framework still in legislative process. Limited ANVISA capacity for AI device evaluation. Digital health infrastructure gaps in underserved reg... Notable: LGPD is Latin America's most comprehensive data protection law. Active AI legislation process. Growing health AI startup ecosystem in São Paulo.

UAE Middle East; Emerging; overall 4.6/10; MOH (Ministry of Health and Prevention), HAAD, DHA; 35 devices; first AI instrument 2017. Scores: P5 C5 A5 T5 E5 PM4 L3. Privacy: Federal Decree-Law No. 45/2021 (Personal Data Protection). AI rule: National AI Strategy 2031, Minister of State for AI appointed (2017). Challenges: Multi-authority regulatory landscape (federal and emirate-level). Rapid strategy

ambitions vs. regulatory maturity. Building domestic healthcare data infrast... Notable: First country to appoint a Minister of State for AI (2017). Ambitious National AI Strategy 2031. Hub for health AI startups and international collaborations.

Saudi Arabia Middle East; Emerging; overall 4.4/10; SFDA (Saudi Food and Drug Authority); 25 devices; first AI instrument 2023. Scores: P5 C4 A5 T5 E5 PM4 L3. Privacy: PDPL (Personal Data Protection Law, 2023). AI rule: SDAIA AI Ethics Principles (2023), National AI Strategy. Challenges: Regulatory framework still maturing. Building domestic regulatory expertise. Dependency on international recognition pathways. Limited public health data inf... Notable: Massive AI investment under Vision 2030. SDAIA established as dedicated AI authority. NEOM project integrating AI healthcare. Rapid digital health transforma...

India Asia; Developing; overall 4.3/10; CDSCO (Central Drugs Standard Control Organisation); 30 devices; first AI instrument 2023. Scores: P5 C4 A4 T4 E5 PM4 L4. Privacy: Digital Personal Data Protection Act (DPDP, 2023). AI rule: No AI-specific law. NITI Aayog AI Strategy and Responsible AI guidelines (2021).. Challenges: Regulatory framework still maturing for AI devices. Limited CDSCO capacity for AI/ML evaluation. Infrastructure gaps in digital health. Balancing innovation ... Notable: DPDP Act 2023 is India's first comprehensive data protection law. Ayushman Bharat Digital Mission building foundational digital health infrastructure. IndiaA...

Thailand Asia; Developing; overall 3.9/10; Thai FDA (Food and Drug Administration); 15 devices; first AI instrument 2022. Scores: P5 C4 A4 T4 E4 PM3 L3. Privacy: PDPA (Personal Data Protection Act, 2022). AI rule: National AI Strategy and Action Plan (2022-2027). No binding AI law.. Challenges: Regulatory framework for AI devices still developing. Digital health infrastructure building. Balancing medical tourism innovation with domestic healthcare p... Notable: PDPA implementation is a major privacy milestone. Thailand 4.0 drives digital health investment. Medical tourism sector catalyzing AI adoption.

South Africa Africa; Emerging; overall 3.6/10; SAHPRA (South African Health Products Regulatory Authority); 10 devices; first AI instrument 2021. Scores: P6 C3 A3 T3 E4 PM3 L3. Privacy: POPIA (Protection of Personal Information Act, 2021). AI rule: No AI-specific regulation. AI Policy Framework under development.. Challenges: SAHPRA capacity and resource constraints. Digital infrastructure gaps. Brain drain in health AI expertise. Balancing innovation with addressing basic healthc... Notable: POPIA is Africa's most comprehensive data protection law. CSIR and universities driving AI health research. Growing health tech ecosystem in Cape Town and Jo...

Kenya Africa; Early; overall 2.7/10; PPB (Pharmacy and Poisons Board); 3 devices; first AI instrument 2019. Scores: P5 C2 A3 T2 E3 PM2 L2. Privacy: Data Protection Act (2019). AI rule: No AI-specific regulation. Blockchain and AI Taskforce Report (2019).. Challenges: Limited regulatory capacity. Infrastructure constraints. Prioritizing basic healthcare access alongside innovation. Dependency on international standards and... Notable: Kenya is a mobile health innovation leader in Africa. Data Protection Act among earliest in East Africa. Growing AI-for-health research at universities and t...

Nigeria Africa; Early; overall 2.6/10; NAFDAC (National Agency for Food and Drug Administration and Control); 5 devices; first AI instrument 2023. Scores: P4 C2 A3 T2 E3 PM2 L2. Privacy: NDPR (Nigeria Data Protection Regulation, 2019) / NDPA (Nigeria Data Protection Act, 2023). AI rule: National AI Strategy (draft). No binding AI regulation.. Challenges: Limited regulatory capacity for AI devices. Digital infrastructure gaps. Healthcare access challenges precede AI adoption. Funding constraints for regulatory... Notable: NDPA 2023 is a significant step in data governance. Growing AI research community (e.g., Data Science Nigeria). Mobile health innovations bypassing tradition...

3.8 Global trends

Table 3.5 records structured trend objects with adoption level, emergence year, and narrative description. Convergence does not mean identical laws; it means shared principles and similar instruments (risk-based SaMD classification, GDPR-style protection, GMLP, lifecycle / PCCP thinking, and a shift from voluntary ethics toward binding rules such as the EU AI Act).

Table 3.5: Global trends in AI healthcare regulation

Trend	Description	Adoption	Since
Convergence on Risk-Based Regulation	Most jurisdictions adopting risk-based classification of AI medical devices, influenced by IMDRF framework.	High	2017
Comprehensive Data Protection Laws	Post-GDPR wave of national data protection laws with health data as sensitive category.	High	2018
AI-Specific Legislation	Growing trend toward dedicated AI laws (EU AI Act leading, followed by Brazil, Canada, South Korea proposals).	Medium	2021
International Harmonization Efforts	IMDRF, GMLP (FDA-Health Canada-MHRA), and WHO guidance driving convergence.	Medium	2019
Ethical AI Frameworks	Proliferation of national AI ethics principles, mostly voluntary, increasingly being legislated.	High	2018
Regulatory Sandboxes for Health AI	Controlled testing environments for innovative AI devices before full regulatory compliance.	Low	2020
Real-World Evidence Acceptance	Growing regulatory acceptance of real-world data and evidence for AI device evaluation.	Medium	2019
Adaptive Algorithm Governance	Frameworks for managing continuously learning AI systems (e.g., FDA Predetermined Change Control Plan).	Low	2021
AI Adoption in Radiology and Digital Pathology Workflows	AI image analysis is increasingly deployed for triage and decision support in radiology and pathology, driving demand for clinical validation, transparency for clinicians, and post-deployment monitoring.	High	2021
Genomic AI for Precision Medicine (Data Governance Focus)	Genomic interpretation and clinical decision support are expanding, with governance emphasizing privacy, representative datasets, and controlled cross-border data use.	Medium	2020
AI-Augmented Drug Discovery and Clinical Development Support	AI supports discovery and development workflows and increasingly influences clinical evidence generation, increasing focus on documentation, auditability, and liability assumptions.	Medium	2021
Hospital Deployment of Clinical Decision Support (Workflow Integration)	Hospitals and clinics integrate AI into electronic workflows, making Total Product Lifecycle governance and user training central to safe deployment.	High	2022
Public Health Surveillance and Outbreak Support with AI	AI systems are being explored for population-level surveillance and outbreak support, requiring accountability mechanisms because impacts extend beyond individual patients.	Medium	2020

3.9 Clinical use cases and milestones

Use-case readiness in the dashboard is a weighted mean of theme scores (radiology weights clinical validation and post-market; genomics weights privacy). Imaging SaMD therefore maps naturally to US/EU clinical-validation strength; genomics maps to GDPR-aligned privacy. Hospital CDS emphasises Total Product Lifecycle thinking [13], [26]; public-health AI requires accountability beyond individual patients [22]; foundation models still lack settled SaMD templates.

Key milestones shaping the score patterns include the EU AI Act (2024), FDA AI/ML SaMD Action Plan (2021) and PCCP guidance (2023), WHO ethics guidance (2021), GMLP (2021), and national data-protection waves (PIPL, DPDP, POPIA, NDPA, LGPD).

4. Conclusions

4.1 Summary of findings

The global regulatory landscape for AI in healthcare is characterised by rapid evolution, converging principles, and persistent diversity. The comparative results in Chapter 3 support five condensed claims:

1. **Risk-based SaMD pathways and data-protection laws are the most mature themes globally**, which is why approval process and privacy lead the theme averages.
2. **Europe and North America dominate the overall ranking**, with the EU/Germany strongest on enforceable ethics and transparency, and the United States strongest on clinical validation and device volume.
3. **Asia is internally heterogeneous**: Japan, Singapore, South Korea, and China are Advanced, while India and Thailand remain Developing—so regional averages hide strategy-critical differences.
4. **Lifecycle governance lags**: post-market surveillance and liability show lower averages and large gaps, especially between Advanced and Early jurisdictions.
5. **Peer gaps are theme-specific**: the US–EU–China–India–South Africa comparison shows Advanced leaders dominate most themes, while Developing/Emerging peers lag most on clinical validation, post-market, and liability.

Lifecycle governance—post-market surveillance, predetermined change control, and liability for adaptive algorithms—is the least mature cluster of themes and the most important frontier as clinical AI moves from static models to continuously updated systems [12], [13].

4.2 Implications for stakeholders

Developers: Use the theme radar and pairwise comparisons to choose first markets by product risk: imaging SaMD may prioritise US/EU clinical-validation strength; genomics products may prioritise GDPR-aligned privacy; adaptive algorithms need PCCP-like change control.

Policymakers: Gaps on clinical validation and post-market themes identify where guidance and inspectorate capacity yield the largest safety return. Emerging jurisdictions can leapfrog by adopting IMDRF/GMLP templates rather than inventing bespoke scorecards.

Researchers and students: The open dataset and dashboard make comparative AI governance reproducible and teachable; the country profiles in this report are a static snapshot of the same underlying JSON.

4.3 Recommendations

1. Expand IMDRF/WHO coordination for AI-specific clinical-evidence methods;
2. Build regulatory capacity in developing countries via technical assistance and mutual-recognition pilots;
3. Operationalise ethics through measurable fairness, monitoring, and documentation standards;
4. Update liability frameworks before harm cases proliferate;

5. Maintain public dashboards and open datasets for transparent comparative governance research;
6. Treat foundation-model medical products as a dedicated evaluation track rather than forcing them into static SaMD templates alone.

These recommendations synthesise WHO, IMDRF, FDA/Health Canada/MHRA GMLP, the EU AI Act, NIST AI RMF, and the peer-reviewed literature cited in this report [7], [8], [15], [22], [28]; they are not the output of a new Delphi survey.

4.4 Contributions

This project contributes (i) a coded 20-country, seven-theme dataset; (ii) an explicit scoring and maturity method with documented gap and readiness formulae; (iii) comparative figures and country profiles suitable for academic presentation; (iv) an interactive dashboard for exploration; and (v) this report linking problem, literature, method, and results.

4.5 Limitations and future work

Scores are expert syntheses, not official ratings, and will drift as laws change. Device-approval counts are incomplete where agencies do not publish AI-specific tallies. The sample over-represents Advanced jurisdictions relative to the true global distribution of countries. Regional means for Oceania and South America rest on single-country observations (Australia, Brazil) and should not be over-generalised.

Future work may incorporate live regulatory feeds, formal expert Delphi scoring, expanded country coverage (especially Latin America and additional African regulators), and dedicated evaluation of foundation-model medical products.

Success in governing AI healthcare will depend on balancing innovation with safety, strengthening international harmonisation, and ensuring frameworks adapt to foundation models, continuous learning, and equitable access across regions.

Data sources

Primary regulatory and policy sources used in dataset curation include the FDA AI/ML SaMD Action Plan and PCCP guidance; EU AI Act, MDR, and GDPR texts; WHO *Ethics and Governance of AI for Health*; OECD.AI Policy Observatory; IMDRF SaMD framework; GMLP guiding principles; NIST AI Risk Management Framework; and EMA, MHRA, PMDA, NMPA, and national AI strategy documents.

References

- [1] E. J. Topol, “High-performance medicine: The convergence of human and artificial intelligence,” *Nature Medicine*, vol. 25, pp. 44–56, 2019.
- [2] P. Rajpurkar, E. Chen, O. Banerjee, and E. J. Topol, “Ai in health and medicine,” *Nature Medicine*, vol. 28, pp. 31–38, 2022.
- [3] U.S. Food and Drug Administration, *Artificial intelligence/machine learning (ai/ml)-based software as a medical device (samd) action plan*, <https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-and-machine-learning-software-medical-device>, 2021.
- [4] E. Wu, K. Wu, R. Daneshjou, et al., “How medical AI devices are evaluated: Limitations and recommendations from an analysis of FDA approvals,” *Nature Medicine*, vol. 27, pp. 582–584, 2021.
- [5] S. Gerke, T. Minssen, and I. G. Cohen, “Ethical and legal challenges of artificial intelligence-driven healthcare,” *Artificial Intelligence in Healthcare*, pp. 295–336, 2020.
- [6] D. S. Char, N. H. Shah, and D. Magnus, “Implementing machine learning in health care — addressing ethical challenges,” *New England Journal of Medicine*, vol. 378, no. 11, pp. 981–983, 2018.
- [7] International Medical Device Regulators Forum, *Software as a medical device: Possible framework for risk categorization and corresponding considerations*, 2014.
- [8] European Parliament and Council, *Regulation (eu) 2024/1689 — artificial intelligence act*, 2024.
- [9] Pharmaceuticals and Medical Devices Agency (Japan), *Dash for samd: Regulatory framework*, 2020.
- [10] National Medical Products Administration (China), *Guiding principles for registration review of AI medical device software*, 2021.
- [11] U. J. Muehlematter, P. Daniore, and K. N. Vokinger, “Approval of artificial intelligence and machine learning-based medical devices in the USA and Europe (2015–20),” *The Lancet Digital Health*, vol. 3, no. 3, e195–e203, 2021.
- [12] B. Babić, S. Gerke, T. Evgeniou, and I. G. Cohen, “Algorithms on regulatory lockdown in medicine,” *Science*, vol. 366, no. 6470, pp. 1202–1204, 2019.
- [13] T. J. Hwang, A. S. Kesselheim, and K. N. Vokinger, “Lifecycle regulation of AI- and ML-based software devices in medicine,” *JAMA*, vol. 322, no. 23, pp. 2285–2286, 2019.
- [14] U.S. Food and Drug Administration, *Marketing submission recommendations for a pre-determined change control plan for ai/ml-enabled device software functions*, 2023.
- [15] FDA, Health Canada, and MHRA, *Good machine learning practice for medical device development: Guiding principles*, 2021.
- [16] B. Goodman and S. Flaxman, “European union regulations on algorithmic decision-making and a “right to explanation”,” *AI Magazine*, vol. 38, no. 3, pp. 50–57, 2017.
- [17] I. G. Cohen and M. M. Mello, “HIPAA and protecting health information in the 21st century,” *JAMA*, vol. 320, no. 3, pp. 231–232, 2018.

- [18] Z. Obermeyer, B. Powers, C. Vogeli, and S. Mullainathan, “Dissecting racial bias in an algorithm used to manage the health of populations,” *Science*, vol. 366, no. 6464, pp. 447–453, 2019.
- [19] L. Seyyed-Kalantari, H. Zhang, M. B. A. McDermott, I. Y. Chen, and M. Ghassemi, “Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in under-served patient populations,” *Nature Medicine*, vol. 27, pp. 2176–2182, 2021.
- [20] M. Ghassemi, L. Oakden-Rayner, and A. L. Beam, “The false hope of current approaches to explainable AI in health care,” *The Lancet Digital Health*, vol. 3, no. 11, e745–e750, 2021.
- [21] C. Rudin, “Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead,” *Nature Machine Intelligence*, vol. 1, no. 5, pp. 206–215, 2019.
- [22] World Health Organization, *Ethics and governance of artificial intelligence for health*, Geneva, 2021.
- [23] A. Jobin, M. Ienca, and E. Vayena, “The global landscape of AI ethics guidelines,” *Nature Machine Intelligence*, vol. 1, no. 9, pp. 389–399, 2019.
- [24] B. Mittelstadt, “Principles alone cannot guarantee ethical AI,” *Nature Machine Intelligence*, vol. 1, no. 11, pp. 501–507, 2019.
- [25] J. Morley, L. Floridi, L. Kinsey, and A. Elhalal, “From what to how: An initial review of publicly available AI ethics tools, methods and research,” *Science and Engineering Ethics*, vol. 26, pp. 2141–2168, 2020.
- [26] C. Petersen, J. Smith, R. R. Freimuth, et al., “Recommendations for the safe, effective use of adaptive CDS in the US healthcare system,” *JAMIA*, vol. 29, no. 4, pp. 677–684, 2022.
- [27] W. N. Price, S. Gerke, and I. G. Cohen, “Potential liability for physicians using artificial intelligence,” *JAMA*, vol. 322, no. 18, pp. 1765–1766, 2019.
- [28] National Institute of Standards and Technology, *Artificial intelligence risk management framework (AI RMF 1.0)*, 2023.